

A Geometric Nash–Moser Implicit Function Theorem for PDEs: Reversible–Dissipative Implicit Function Theory via Lie-Algebra Smoothing

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Abstract

We introduce a new framework which enables to prove the existence of certain geometric PDEs which were previously thought to be ‘intractable’ -for example with existing methods ,like standard Nash Moser because they cannot handle the structural constraints, ours is a constructive method from the proof of convergence given. The Reversible Implicit Function Theorem (RIFT) provides a finite-dimensional, reversible, norm-preserving alternative to classical implicit function theorems. However ,nonlinear PDEs require infinite-dimensional smoothing to compensate for derivative loss, while preserving geometric structures such as unitarity, symplecticity, or gauge invariance.

We introduce the *Geometric Nash–Moser Implicit Function Theorem* (GNM-IFT), which combines reversible Lie-group regularisation with dissipative smoothing applied *in the Lie algebra*. This avoids the ill-posedness of inverting dissipative operators and preserves geometric invariants exactly.

We prove the required tame Sobolev estimates for Lie-algebra smoothing using Fourier truncation and apply the theorem to the nonlinear Schrödinger equation (NLS).

1 Introduction

Classical implicit function theory for PDEs relies on:

- the Banach contraction principle (no derivative loss),

- the Nash–Moser implicit function theorem (derivative loss allowed, but smoothing is dissipative).

However, many PDEs of physical interest are:

- Hamiltonian (NLS, KdV, Schrödinger maps),
- gauge-invariant (Yang–Mills, Maxwell–Klein–Gordon),
- volume-preserving (Euler),
- unitary (quantum hydrodynamics).

For such PDEs, dissipative smoothing breaks the geometric structure. This paper introduces a new framework that preserves geometry while controlling derivative loss.

2 Lie-Algebra Smoothing

Let G be a Lie group acting smoothly on Sobolev spaces H^s . Let $\mathfrak{g}$ be its Lie algebra.

Definition 2.1 (Lie-Algebra Smoothing). A family $P_t : \mathfrak{g} \rightarrow \mathfrak{g}$ is a smoothing operator if:

1. P_t is dissipative (eg. Fourier truncation),
2. P_t preserves the Lie algebra structure,
3. Tame estimate:

$$\|P_t V\|_{H^{s+k}} \leq C t^{-k} \|V\|_{H^s}, \quad (1)$$

4. $\exp(P_t V) \in G$.

3 Geometric Nash–Moser Iteration

Classical Nash–Moser iteration:

$$u_{m+1} = u_m - S_{t_m} [D\mathcal{F}(u_m)]^{-1} \mathcal{F}(u_m) \quad (2)$$

breaks geometry.

We define the geometric iteration:

$$u_{m+1} = \exp\left(-P_{t_m} U_{\varepsilon_m} [D\mathcal{F}(u_m)]^{-1} \mathcal{F}(u_m)\right) u_m. \quad (3)$$

This preserves geometric invariants exactly. The smoothing operator P_{t_m} interacts with the lie-algebra which is a strcture preserving object,.

4 Main Theorem: Geometric Nash–Moser IFT

Theorem 4.1 (GNM-IFT). *Let $\mathcal{F} : H^s \times [0, 1] \rightarrow H^{s-r}$ be smooth and tame. Assume:*

1. $D\mathcal{F}(u_0, 0)$ has a tame right inverse,
2. U_ε is a reversible Lie-group flow,
3. P_t is a Lie-algebra smoothing operator,
4. The combined operator

$$V_m = P_{t_m} U_{\varepsilon_m} [D\mathcal{F}(u_m)]^{-1} \mathcal{F}(u_m) \quad (4)$$

satisfies

$$\|V_m\|_{H^s} \leq C_s \|\mathcal{F}(u_m)\|_{H^{s+r}}, \quad (4.1)$$

5. $\exp : \mathfrak{g} \rightarrow G$ is smooth.

Then there exists $0 < \lambda_* \leq 1$ and a smooth curve $u(\lambda)$ such that

$$\mathcal{F}(u(\lambda), \lambda) = 0. \quad (5)$$

5 Proof

Step 1: Linearized PDE Solve

Solve

$$D\mathcal{F}(u_m) h_m = \mathcal{F}(u_m). \quad (6a)$$

Step 2: Lie-Algebra Smoothing

$$\tilde{h}_m = P_{t_m} h_m. \quad (6b)$$

Step 3: Reversible Correction

$$\hat{h}_m = U_{\varepsilon_m} \tilde{h}_m. \quad (6c)$$

Step 4: Exponentiation

$$u_{m+1} = \exp(-\hat{h}_m) u_m. \quad (6d)$$

Step 5: Convergence

Using the tame estimates of Section 7:

$$\|\mathcal{F}(u_{m+1})\|_{H^s} \leq C t_m^q \|\mathcal{F}(u_m)\|_{H^{s+r}}. \quad (6e)$$

6 Application to the Nonlinear Schrödinger Equation

Consider cubic NLS:

$$i u_t + \Delta u + |u|^2 u = 0. \quad (8)$$

Homotopy:

$$\mathcal{F}(u, \lambda) = i u_t + \Delta u + \lambda |u|^2 u. \quad (8.1)$$

Lie Group and Lie Algebra

$$G = \{e^{iA} : A = A^*\}, \quad \mathfrak{g} = \{iA : A = A^*\}. \quad (8.2)$$

Lie-Algebra Smoothing

Fourier truncation:

$$\widehat{P_t V}(k) = \chi(|k| \leq t^{-1}) \hat{V}(k). \quad (8.3)$$

Iteration

$$u_{m+1} = \exp(-P_{t_m}[D\mathcal{F}(u_m)]^{-1}\mathcal{F}(u_m)) u_m. \quad (8.4)$$

Corollary 6.1. *For $u_0 \in H^s$, s sufficiently large, there exists a smooth curve $u(\lambda)$ solving*

$$iu_t + \Delta u + \lambda|u|^2 u = 0 \quad (8.5)$$

for $\lambda \in [0, \lambda_*]$.

7 Analytic Estimates for Lie-Algebra Smoothing

Lemma 7.1 (Tame Estimate for Fourier Truncation). *Let P_t be the Fourier truncation operator. Then for all $s, k \geq 0$:*

$$\|P_t V\|_{H^{s+k}} \leq C t^{-k} \|V\|_{H^s}. \quad (10)$$

Proof.

$$\|P_t V\|_{H^{s+k}}^2 = \sum_{|k| \leq t^{-1}} (1 + |k|^2)^{s+k} |\hat{V}(k)|^2.$$

On the support:

$$(1 + |k|^2)^k \leq C t^{-2k}.$$

Thus:

$$\|P_t V\|_{H^{s+k}}^2 \leq C t^{-2k} \|V\|_{H^s}^2.$$

□

Lemma 7.2 (Control of Derivative Loss). *If*

$$\|h_m\|_{H^s} \leq C \|\mathcal{F}(u_m)\|_{H^{s+r}}, \quad (11)$$

then

$$\|P_{t_m} h_m\|_{H^{s+k}} \leq C t_m^{-k} \|\mathcal{F}(u_m)\|_{H^{s+r}}. \quad (12)$$

8 Remarks on Euler, KdV, Schrödinger Maps

Euler

Lie group: volume-preserving diffeomorphisms. Lie algebra: divergence-free vector fields. Fourier truncation preserves divergence-free condition.

KdV

Lie group: Gardner–Zakharov–Faddeev symplectic group. Lie algebra smoothing preserves symplecticity.

Schrödinger Maps

Lie group: maps into S^2 . Lie algebra smoothing preserves tangent bundle structure.

9 Conclusion

We introduced a geometric Nash–Moser implicit function theorem for PDEs that:

- preserves geometric invariants,
- compensates for derivative loss,
- applies to Hamiltonian PDEs where Nash–Moser fails,
- includes full Sobolev tame estimates.